Higher National Diploma in Engineering (Electrical)

First Year, Second Semester Examination - 2019

EE 1209-Electronic Circuits & Systems I

Answer any 5 questions Time : 3 hours

Q1.

(i). Briefly explain the **majority and minority carriers** of **p and n type semiconductors**. [03 marks] **(ii).** Explain the operation of a **semiconductor diode** during **forward and reverse biased conditions**. [03 marks] **(iii).** Draw the **waveform of the output voltage** V_0 for Figure Q1a by considering that the **diode is ideal**, $V_{\text{in}} = V_m \sin \omega t$ and $V < V_m$. [05 marks] **(iv).** A **10 V Zener diode** is used as a **voltage regulator** & load resistance is $\mathbf{3\ k\Omega}$ as shown in Figure Q1b . The series resistor $\mathbf{R}$ is $\mathbf{1\ k\Omega}$, and the input voltage $\mathbf{V_i}$ is $\mathbf{16\ V}$. Find the following: a). The **load voltage** b). The **Zener current** c). The **power dissipated in the Zener diode** d). Explain the **operation of the circuit as a voltage regulator** with different loads for the same input. [09 marks]

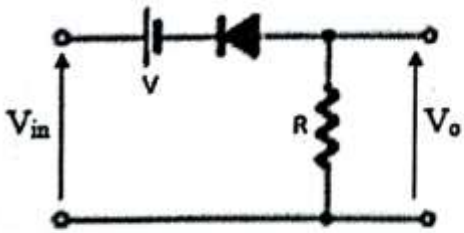

Figure Q1a

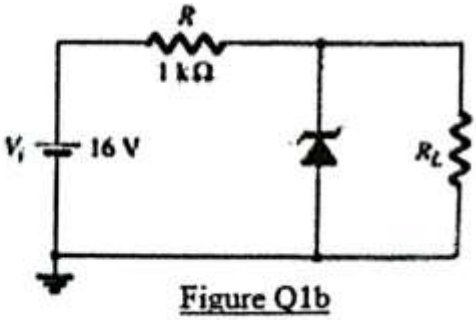

Figure Q1b

Q2.

(i). Explain the **importance of the peak inverse voltage** of the diode. [03 marks] **(ii).** The **bridge rectifier** shown in Figure Q2a has four ideal diodes and is connected to $\mathbf{R = 2\ k\Omega}$ as load resistance. The supply voltage is $\mathbf{10 \sin 300t}$. Calculate: a). the **output dc voltage** b). **Efficiency of the circuit** c). **PIV of the diode** [10 marks] **(iii).** The **voltage regulator circuit** is modified as shown in Figure Q2b . a). Sketch the **waveform of the output voltage (V_0)** for the same input voltage. b). Calculate the **output DC voltage and PIV**. [07 marks]

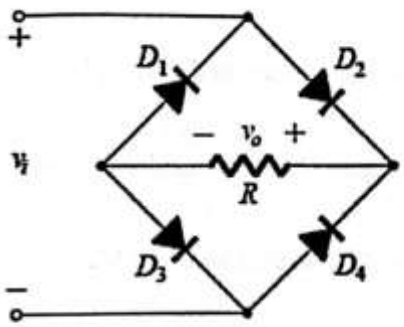

Figure Q2a

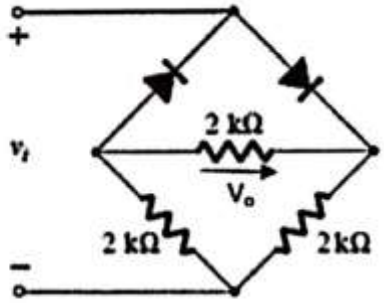

Figure Q2b

Q3.

(i). Explain why $\mathbf{V_{CE}}$ is usually chosen around $\mathbf{V_{CC}/2}$ in amplifiers. [02 marks] **(ii).** A **CE amplifier circuit** is shown in Figure Q3 . A **Si transistor** is used with $\beta = \mathbf{100}$. a). Find the **DC operating point** of the amplifier. b). Draw the **AC equivalent circuit** considering very high input resistance of the transistor. c). Calculate **voltage gain, current gain & input resistance**. [13 marks] **(iii).** Draw the **AC equivalent circuit** by **removing the C_2 capacitor** in Figure Q3 & calculate the **voltage gain**. [05 marks]

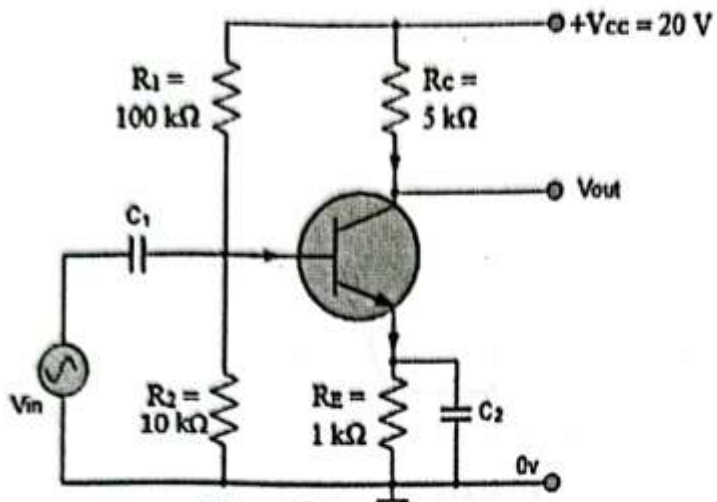

Figure Q3

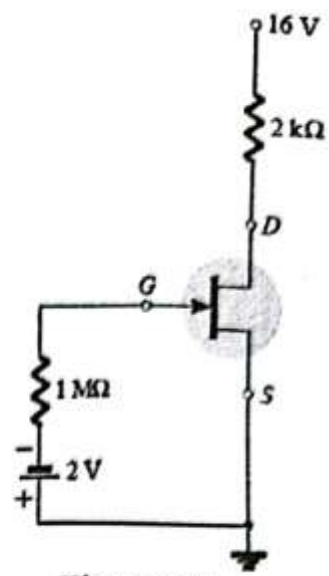

Figure Q4

Q4.

(i). Explain briefly the **necessity of negative V_{GS} voltage** to the operation of **JFET**. [04 marks] **(ii).** a. Define **transconductance (g_m)** for JFET and derive the expression for g_m with the terms I_D, V_P, and I_{DSS}. b. A JFET has parameters $\mathbf{I_{DSS} = 5\ mA}$, $\mathbf{V_P = -2\ V}$. Find the value of **drain current (I_D)** at $\mathbf{V_{GS} = -1.5\ V}$. [08 marks] **(iii).** A **fixed biased JFET amplifier** having parameters $\mathbf{I_{DSS} = 10\ mA}$ & $\mathbf{V_P = -8\ V}$ is shown in Figure Q4 . Determine $\mathbf{V_{GS}}$, $\mathbf{I_D}$ and $\mathbf{V_{DS}}$. [08 marks]

Q5.

(i). Explain the difference between a **combinational logic circuit and a sequential logic circuit** . [02 marks] **(ii).** Convert the following numbers with the indicated bases to **binary**: a. $\mathbf{36.375_{10}}$ b. $\mathbf{374.2_8}$ c. $\mathbf{17E.F6_{16}}$ [06 marks] **(iii).** Find the values of the following using the **8-bit binary form of the 2's complement method**: a. $\mathbf{37 - 18}$ b. $\mathbf{-80 - 32}$ [06 marks] **(iv).** A logic circuit is shown in Figure Q5 has four inputs $\mathbf{A, B, C, D}$ and the output is $\mathbf{Y}$. a. Draw the **truth table** b. Write a **Boolean expression for output** c. **Simplify the expression** [06 marks]

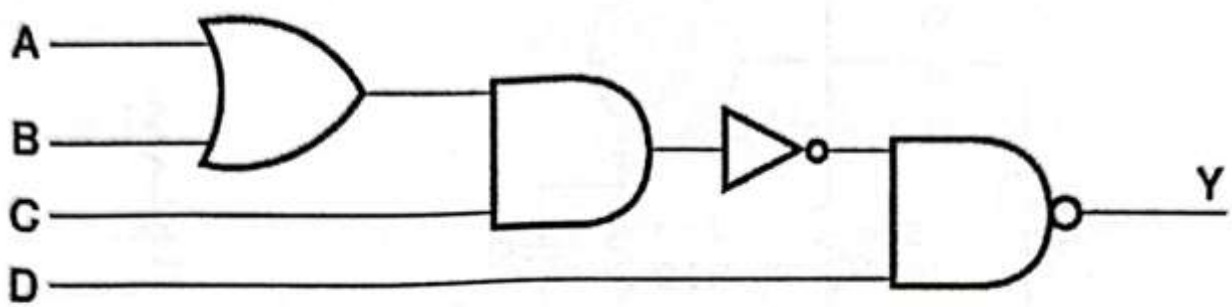

Figure Q5

Q6.

(i). State **de Morgan's theorem** for two logic variables $\mathbf{X}$ and $\mathbf{Y}$. [02 marks] **(ii).** Draw a **logic diagram** for the function $\mathbf{F = (A' + B')(C + D)}$ using: a. only **NOR gates** b. only **NAND gates** [06 marks] **(iii).** Use a **Karnaugh map** to simplify the following: a. $\mathbf{X = (AB + A'C + ABC)}$ b. $\mathbf{Y = (0, 1, 2, 5, 6, 7, 9, 11, 12, 14)}$ (The values are minterms for a four-variable map) [06 marks] **(iv).** Write **short notes** on the following: a. **Logic gates** b. **Karnaugh map** c. **Redundant group** [06 marks]